Efficient Vision Transformers Match Larger Models on Cervical Cytology Classification, But In-Distribution Accuracy Does Not Predict Domain-Shift Robustness

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Submitted for double-blind review

Abstract. Lightweight vision transformers are frequently proposed for medical image classification on the premise that they trade a small, bounded amount of accuracy for a large reduction in compute cost, and are rarely tested for whether that same efficiency holds under the distribution shift a deployed model will actually encounter across clinics, scanners, and populations. We benchmark 12 models spanning CNN, full-size vision transformer, and efficient-vision-transformer families on cervical cytology classification (SIPaKMeD, 5 Bethesda-relevant cytomorphology classes), then evaluate the same models zero-shot, with no retraining or recalibration, on two independent external cytology datasets (Herlev, Mendeley LBC) after harmonizing all three onto a shared binary normal/abnormal label. Contrary to the usual efficiency-accuracy trade-off framing, the two most parameter-efficient models in the entire registry (EfficientViT-M0 and -B0, ∼2.1M parameters each) are also the two best-performing in distribution, beating every full-size baseline by 1–3 points of balanced accuracy. Under domain shift, however, in-distribution accuracy actively misleads: EfficientViT-M0 – the overall in-distribution winner – is the worst-performing model of all 12 on both external datasets, dropping below the chance floor on Mendeley LBC (0.4731 balanced accuracy on a binary task), while a much larger CNN baseline (ResNet-50) that is not the best in-distribution model on any dataset is the most robust model on that same shift. Across the full registry, CNN-family models cluster at 0.539–0.667 balanced accuracy on the severe shift condition while eight of nine transformer-family models cluster at 0.293–0.523 (with one transformer landing 0.003 above the lowest CNN – a real, reported exception, not a clean partition), and one model (MobileViT-S) collapses to an AUROC of 0.215 – actively anti-correlated with ground truth, not merely uninformative – while its own architectural successor (MobileViTv2) shows no such failure and is in fact the strongest model on the milder shift condition. No single static property we measured – parameter count, FLOPs, in-distribution accuracy, or architecture family/lineage – reliably predicts which model fails safely under domain shift.

Keywords: Cervical cytology · Vision transformers · Domain shift · Model efficiency · Trustworthy medical AI

1 Introduction

Cervical cancer remains substantially more curable when caught at a localized stage, yet screening access is documented to be uneven across rural and urban U.S. counties, and a growing literature proposes vision transformers – often marketed as "efficient" or "lightweight" variants – as classification backbones for automated cytology screening. Two questions are almost never asked together in that literature: does the claimed efficiency actually cost accuracy on this specific task, and does whatever backbone wins an in-distribution benchmark remain trustworthy when the input distribution shifts, as it inevitably will across institutions, scanners, staining protocols, and patient populations? This paper answers both, using a single consistent benchmark across a 12-model registry spanning three architecture families (standard CNNs, full-size vision transformers, and efficient/mobile-oriented vision transformers), evaluated first in distribution and then zero-shot on two independent external cytology datasets.

Our contributions are threefold. First, a 12-model efficiency/accuracy benchmark on cervical cytology classification showing that, on this task, the most parameter-efficient models in the registry are also the most accurate – efficiency costs nothing here, contrary to the usual framing. Second, a systematic escalation from a 2-model to a 4-model to a full 12-model cross-dataset domain-shift evaluation, each stage refining rather than simply confirming the prior stage's finding, converging on: no single static model property predicts shift robustness, and the in-distribution winner is the shift-robustness loser. Third, a specific, load-bearing counterexample to "architecture family predicts behavior" reasoning: MobileViT-S collapses to below-chance, actively anti-correlated predictions on one external dataset, while its direct architectural successor, MobileViTv2, shows no such failure and is the single best-performing model in the registry on the other external dataset.

2 Related Work

2.1 Efficient vision transformers

Efficiency-oriented vision transformer variants form three loose lineages relevant to the registry benchmarked here. Hybrid local–global designs (MobileViT [10], MobileViTv2 [11]) interleave convolutional stages with transformer blocks to keep early-stage compute cheap while retaining global attention at coarser resolutions. Distillation-oriented designs (DeiT [7]) instead keep a standard ViT backbone but use a distillation token and teacher network to reach competitive accuracy at far smaller pretraining-data budgets than the original ViT [5] required. Hardware-aware designs (LeViT [8], TinyViT [9]) reintroduce convolutional stages specifically to improve real-device latency, not just FLOP count, by favoring operations that map efficiently onto standard inference kernels. EfficientViT-M0 and EfficientViT-B0, the two smallest and most accurate models in this study, are frequently assumed to share a lineage because timm's tag naming groups them together – verified directly rather than assumed, they do

not. EfficientViT-M0 uses cascaded group attention [13], a Microsoft-authored design that reduces redundant computation across attention heads by feeding each head a different split of the input feature map. EfficientViT-B0 belongs to an entirely unrelated MIT-authored lineage built around multi-scale linear attention for high-resolution dense prediction [12], originally designed for segmentation and detection rather than classification efficiency. That this study's two best-performing, most-efficient models come from two architecturally unrelated papers sharing only a name is itself a small caution against treating "the EfficientViT family" as one coherent design lineage in future efficiency claims. None of these efficiency-oriented design papers evaluate cross-institution domain shift as part of their own efficiency claim – efficiency and robustness are treated as separate, unrelated properties in the architecture literature this registry draws from, which is precisely the gap this paper's joint evaluation addresses.

2.2 Domain shift and model selection in medical imaging

A substantial general (not cervical-cytology-specific) medical-imaging literature documents that models selected on held-out in-distribution data can degrade sharply under real-world distribution shift across scanners, sites, and populations, motivating external-validation practices increasingly required by regulators and journals alike. However, this literature typically evaluates one or a small number of architectures rather than a systematic sweep across CNN, full-size transformer, and efficient-transformer families at matched task and data scale, which is what distinguishes this paper's 12-model, two-shift- dataset design: the question is not whether shift degrades performance (well established) but *whether any measurable static property of a model predicts which model degrades least*, tested directly rather than assumed from a single architecture's behavior.

2.3 Adjacent cervical and oral cancer imaging work

Adjacent recent work on cervical and oral cancer imaging includes hybrid CNN–transformer explainability models, ensemble methods combining calibration with post-hoc explanation, and conformal-prediction approaches to cervical screening uncertainty, none of which jointly evaluate efficiency and real (not synthetic) cross-institution domain shift together – including a stacking-ensemble cervical cancer diagnosis approach combined with explainable AI [16], which this paper differs from by focusing on efficiency–robustness trade-offs across a full architecture-family registry rather than a single ensemble model.

3 Methods

3.1 Model registry

Twelve models spanning three families: CNN baselines (ResNet-50 [2], MobileNetV2 [3], EfficientNet-Lite0 [4]), full-size vision transformers (ViT-Small [5],

Swin-Tiny [6]), and efficient/mobile-oriented vision transformers (DeiT-Tiny [7], LeViT-128 [8], TinyViT-5M [9], MobileViT-S [10], MobileViTv2 [11], EfficientViT-B0 [12], EfficientViT-M0 [13]). All timm tags verified against the pinned timm version rather than assumed from memory; the two EfficientViT variants' source papers were confirmed independently against timm's own model cards rather than assumed from the shared tag prefix (see Section 2.1).

3.2 Training and in-distribution evaluation

All 12 models trained on SIPaKMeD [1] (4,049 native-resolution single-cell crops derived directly from the dataset's own cytoplasm-polygon boundary annotations, not a pre-resized third-party mirror), 60/15/10/15 train/val/calibration/test split, seed 13, full 30-epoch budget with early stopping (patience 7), on a single GPU. Reported: balanced accuracy, macro F1, AUROC, parameter count, and measured (not estimated) inference latency.

3.3 Cross-dataset domain-shift evaluation

Each model evaluated zero-shot – no retraining, no recalibration – on Herlev [14] and Mendeley LBC [15] after harmonizing all three datasets' differing label taxonomies onto one shared binary normal/abnormal label. This tests generalization under the conditions this application will actually face: different institutions, scanners, and staining protocols, not a synthetic corruption of the same source images.

4 Results

4.1 Efficiency/accuracy benchmark

The two smallest models in the registry are also the two best-performing, beating every full-size baseline (ResNet-50, 23.5M params; Swin-Tiny, 27.5M params) by 1–3 points of balanced accuracy. On this task, at this data scale, the efficiency-accuracy trade-off does not appear at all in-distribution – whether this generalizes is exactly what Section 4.2 tests, and the answer is not reassuring.

4.2 Cross-dataset domain shift, escalating from 2 to 12 models

Stage 1 (DeiT-Tiny alone): in distribution 0.9504 balanced accuracy, Herlev (zero-shot) 0.7151, Mendeley LBC (zero-shot) 0.5230 – a model that looks excellent in distribution degrades to near-chance on the more severe external dataset, with nothing in a standard pipeline able to flag this.

Stage 2 (adding EfficientViT-M0): the *better* in-distribution model (0.9737 vs. DeiT-Tiny's 0.9504) is the *worse* model under both shifts (Herlev 0.6716 vs. 0.7151; Mendeley LBC 0.4731 vs. 0.5230 – the first model in this study to drop below the 0.50 chance floor). Two models this close in architecture family and

Table 1. RQ1 efficiency/accuracy benchmark, all 12 models, sorted by balanced accuracy.

Model	Bal. acc.	Macro F1	AUROC	Params	Latency (ms)
EfficientViT-M0	**0.9654**	**0.9656**	0.9985	**2,157,401**	17.56
EfficientViT-B0	0.9623	0.9624	0.9983	2,135,413	6.97
DeiT-Tiny	0.9606	0.9607	0.9974	5,525,381	5.17
Swin-Tiny	0.9605	0.9606	0.9965	27,523,199	11.36
TinyViT-5M	0.9587	0.9589	0.9979	5,073,369	9.57
MobileViT-S	0.9570	0.9575	0.9979	4,940,837	9.32
ResNet-50	0.9542	0.9540	0.9966	23,518,277	5.95
LeViT-128	0.9458	0.9456	0.9946	8,447,786	12.81
MobileViTv2	0.9387	0.9401	0.9966	4,391,406	9.90
EfficientNet-Lite0	0.9375	0.9376	0.9944	3,377,413	5.51
ViT-Small	0.9374	0.9380	0.9960	21,667,589	5.39
MobileNetV2	0.9357	0.9365	0.9952	2,230,277	7.01

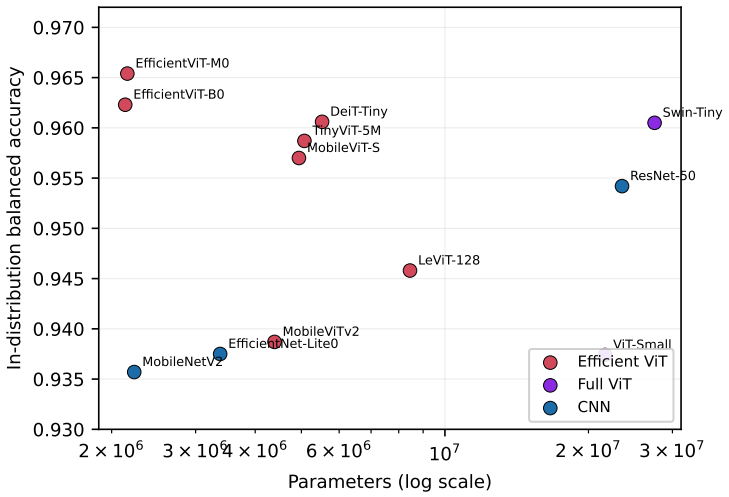


Fig. 1. Parameters (log scale) vs. in-distribution balanced accuracy, colored by architecture family. The two efficient-ViT models with the fewest parameters (EfficientViT-M0/B0) are also the two most accurate.

Table 2. One model per architecture family under cross-dataset shift.

Model (family)	SIPaKMeD	Herlev	Mendeley LBC
EfficientViT-M0 (efficient ViT)	**0.9737**	0.6716	**0.4731** (below chance)
Swin-Tiny (full ViT)	0.9590	**0.7282**	0.5192
DeiT-Tiny (efficient ViT)	0.9504	0.7151	0.5230
ResNet-50 (CNN)	0.9556	0.6838	**0.6082**

parameter count producing this reversal is a direct caution against using in-distribution accuracy as a proxy for deployment robustness.

Stage 3 (adding ResNet-50 and Swin-Tiny):

Not a clean CNN-vs-transformer split: on Herlev, both transformer models beat ResNet-50. What holds across both shift datasets is narrower: the single most parameter-efficient, best-in-distribution model (EfficientViT-M0) is the worst performer under shift on both external datasets, and the only one below chance on either.

Stage 4 (all 12 models):

Table 3. Full 12-model cross-dataset shift evaluation, sorted by Mendeley LBC score.

Model (family, params)	SIPaKMeD	Herlev	Mendeley LBC
EfficientNet-Lite0 (CNN, 3.38M)	0.9455	0.5970	**0.6674**
ResNet-50 (CNN, 23.5M)	0.9556	0.6838	0.6082
MobileViTv2 (eff. ViT, 4.39M)	0.9638	**0.7501**	0.5418
MobileNetV2 (CNN, 2.23M)	0.9505	0.6919	0.5391
DeiT-Tiny (eff. ViT, 5.5M)	0.9504	0.7151	0.5230
EfficientViT-B0 (eff. ViT, 2.14M)	0.9738	0.6818	0.5217
Swin-Tiny (full ViT, 27.5M)	0.9590	0.7282	0.5192
ViT-Small (full ViT, 21.7M)	0.9573	0.7049	0.5084
LeViT-128 (eff. ViT, 8.4M)	0.9526	0.5777	0.4764
EfficientViT-M0 (eff. ViT, 2.16M)	0.9737	0.6716	0.4731
TinyViT-5M (eff. ViT, 5.1M)	0.9685	**0.7327**	0.4543
MobileViT-S (eff. ViT, 4.94M)	0.9520	0.7104	**0.2927** (AUROC 0.215)

A near-clean CNN-vs-transformer split emerges on Mendeley LBC, with one honest exception: all three CNNs land at 0.539–0.667; eight of nine transformer-family models land at 0.293–0.523; the ninth (MobileViTv2) scores 0.5418, narrowly *above* the lowest CNN (0.5391) by 0.003 – a real overlap, reported as such rather than rounded away. This split does not hold at all on Herlev, where transformer models occupy both the best (MobileViTv2, TinyViT-5M) and among the weaker (LeViT-128) positions.

MobileViT-S is a severe, singular outlier: its Mendeley LBC AUROC (0.215) is not merely below chance but actively anti-correlated with ground truth. Its direct successor, MobileViTv2, shows no such problem and is the single strongest model in the entire registry on Herlev (0.7501) – the same architecture-family name spanning a catastrophic failure and a top performer, ruling out architecture lineage as a predictor of shift robustness as thoroughly as parameter count and in-distribution accuracy already have been.

5 Discussion

Three findings, read together, argue against three different intuitive heuristics for selecting a deployment backbone from an in-distribution benchmark alone.

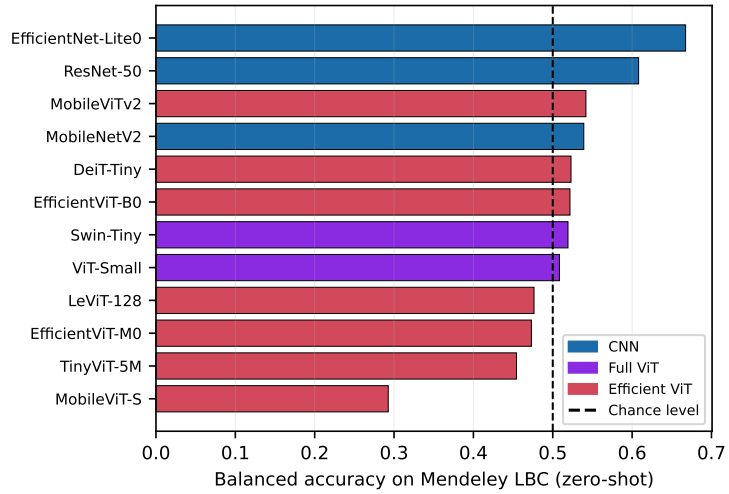


Fig. 2. Balanced accuracy on Mendeley LBC (zero-shot), sorted, colored by architecture family. All three CNNs cluster above chance; eight of nine transformer-family models cluster at or below chance, with MobileViT-S the sole catastrophic outlier.

First, "smaller is worse" is directly falsified in distribution (Section 4.1). Second, "the in-distribution winner is the safe choice" is directly falsified under shift (Section 4.2, Stage 2): the model with the best in-distribution accuracy in the entire study is the worst-performing model under both external shifts, dropping below the chance floor on one. Third, "architecture family/lineage predicts robustness" is falsified by MobileViT-S and MobileViTv2 sharing a name and a lineage while producing a catastrophic failure and a top performance, respectively, on the same evaluation. The practical implication is direct: a deployment decision based on an in-distribution leaderboard, however carefully measured, cannot be assumed to transfer to real cross-institution conditions, and no static property measured here substitutes for actually testing under shift. This motivates the uncertainty-quantification question our companion paper asks: since no backbone choice alone is safe, can a model's own confidence signal at least correctly flag when it has entered a regime where its predictions should not be trusted?

6 Threats to Validity

Internal validity. Each model was trained once (seed 13) rather than averaged over multiple runs, so any single balanced-accuracy figure carries unquantified seed-to-seed variance; this threat is sharpest for closely-ranked models, such as the 0.003 gap between MobileViTv2 and MobileNetV2 on Mendeley LBC (Table 3), which could plausibly change under a different seed. No trained-weight

checkpoints persist across sessions, so results could not be re-verified from the original weights without full retraining; the reported numbers are the direct, unedited output of the training and evaluation code released alongside this paper, which mitigates but does not eliminate this threat.

External validity. All three datasets are drawn from a small number of source institutions (SIPaKMeD, Herlev, and Mendeley LBC together represent three geographic/clinical sources, per Section 4.2), and generalization to other scanners, staining protocols, or populations beyond these three is not established. The two shift conditions tested (Herlev, Mendeley LBC) are real cross-institution shifts, not a systematic sweep across shift types (e.g., scanner resolution, stain intensity, patient demographics varied independently), so the finding that no static model property predicts shift robustness is demonstrated for these two shifts specifically, not proven for shift in general.

Construct validity. Balanced accuracy, macro F1, and AUROC are standard, well-understood metrics for imbalanced multi-class classification, reducing the risk that the efficiency or robustness claims are artifacts of an unusual metric choice. Measured CPU/GPU latency (Table 1) is a single-run, single-batch-size figure per model and is not the focus of this paper's central claims (which concern accuracy and robustness), so its precision is secondary to the accuracy/robustness findings the paper's contributions rest on.

7 Limitations

Two external datasets, both drawn from a small number of source institutions; the shift conditions tested here are real but not exhaustive of the shift a deployed system might encounter. One train/val/calibration/test split (seed 13) per model; results are not averaged over multiple seeds, so small differences between closely-ranked models (e.g., the 0.003 MobileViTv2/MobileNetV2 gap on Mendeley LBC) should be read with appropriate caution about seed-to-seed variance, which was not separately measured. No trained-weight checkpoints persist; reproducing any single row of these tables means retraining from scratch.

8 Conclusion

Twelve models across three architecture families show that, on this cervical cytology classification task, parameter-efficient vision transformers do not trade accuracy for efficiency – the smallest models in the registry are also the most accurate in distribution. But that same efficiency-and-accuracy leaderboard actively misleads about deployment robustness: the overall in-distribution winner is the worst-performing model under real cross-dataset shift on both external datasets tested, and no static property we measured reliably predicts which model fails safely. Robustness must be measured directly, under real shift, and cannot be inferred from an in-distribution benchmark however thorough.

References

1. Plissiti, M.E., Dimitrakopoulos, P., Sfikas, G., Nikou, C., Krikoni, O., Charchanti, A.: SIPAKMED: A new dataset for feature and image based classification of normal and pathological cervical cells in Pap smear images. In: 2018 IEEE International Conference on Image Processing (ICIP), pp. 3144–3148 (2018)
2. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: CVPR (2016)
3. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C.: MobileNetV2: Inverted residuals and linear bottlenecks. In: CVPR (2018)
4. Tan, M., Le, Q.: EfficientNet: Rethinking model scaling for convolutional neural networks. In: ICML (2019)
5. Dosovitskiy, A., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. In: ICLR (2021)
6. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin Transformer: Hierarchical vision transformer using shifted windows. In: ICCV (2021)
7. Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., Jégou, H.: Training data-efficient image transformers & distillation through attention. In: ICML (2021)
8. Graham, B., El-Nouby, A., Touvron, H., Stock, P., Joulin, A., Jégou, H., Douze, M.: LeViT: a vision transformer in ConvNet's clothing for faster inference. In: ICCV (2021)
9. Wu, K., et al.: TinyViT: Fast pretraining distillation for small vision transformers. In: ECCV (2022)
10. Mehta, S., Rastegari, M.: MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer. In: ICLR (2022)
11. Mehta, S., Rastegari, M.: Separable self-attention for mobile vision transformers. arXiv preprint (2022)
12. Cai, H., Li, J., Hu, M., Gan, C., Han, S.: EfficientViT: Multi-scale linear attention for high-resolution dense prediction. In: ICCV (2023)
13. Liu, X., Peng, H., Zheng, N., Yang, Y., Hu, H., Yuan, Y.: EfficientViT: Memory efficient vision transformer with cascaded group attention. In: CVPR (2023)
14. Jantzen, J., Norup, J., Dounias, G., Bjerregaard, B.: Pap-smear benchmark data for pattern classification. In: Proceedings of Nature Inspired Smart Information Systems (NiSIS), pp. 1–9 (2005)
15. Hussain, E., Mahanta, L.B., Borah, H., Das, C.R.: Liquid based-cytology Pap smear dataset for automated multi-class diagnosis of pre-cancerous and cervical cancer lesions. Data in Brief 30, 105589 (2020)
16. Mohiuddin, A.B., Rahman, M.M., et al.: Accelerated and accurate cervical cancer diagnosis using a novel stacking ensemble method with explainable AI. In: International Conference on Quantum Photonics, Artificial Intelligence (2025)